

- e) may be used in manual mode but require qualified operators (that is, traffic controller) – operators with two-way radio can monitor signal performance, warn the roadworks site and manage road users, and
- f) do not use where side roads intersect the roadworks site and are not controlled by a traffic controller or other PTSS.
- g) While PTSS devices may be operated in a few different modes, when using PTSS Type 1 or Type 2 devices, for manually controlled shuttle flow arrangements, they should be used in paired mode using a single HRC whenever possible or suitable. This will prevent a green signal being displayed to both approaches at the same time.

9. Portable boom barriers:

- a) are intended to stop traffic
- b) manage shuttle flow or gating (all stop) operation, and
- c) require qualified operators to operate in manual mode (that is, traffic controller).

When using PTCDs:

- 10. four cones should be placed on the centreline spaced 4 m apart starting from the STOP HERE ON RED SIGNAL or the STOP HERE WHEN DIRECTED sign position (downstream), and
- 11. undertake a risk assessment.

Consider the following when using PTCDs:

- 12. PTCDs must only be manually controlled by qualified operators (that is, traffic controllers).
- 13. They are intended for traffic control of relatively short duration: for roadworks sites that will continue for a longer period without work area location changes, consider installing temporary, rather than portable, traffic signals (see AS 1742.3).
- 14. Where traffic is required to stop, temporary road markings may be installed 6 m in advance of the PTCD to indicate a stop line. The STOP HERE ON RED SIGNAL or STOP HERE WHEN DIRECTED must be installed in accordance with the Queensland MUTCD Part 3.
- 15. Provide warning signs (for example, Signals AHEAD) an appropriate sight distance in advance of any PTCD as shown in Section 2.5.3 (see QGTTM Part 3